

Layer-2 & Switch Attacks

MAC flooding (CAM table overflow)

Send thousands of fake MACs → switch CAM saturates → uninformed ports flood all frames (back to HUB behavior)

Then sniff everything on the LAN

```
macof -i eth0 # dsniff suite MAC flooder
```

VLAN hopping

Double tagging (802.1Q): send a frame with two VLAN tags; switch strips outer, forwards inner to the target VLAN – breaking segmentation

Switch spoofing / DTP: negotiate trunking (DTP) as if you were a switch → join multiple VLANs

```
# set your interface to trunk mode in the lab environment  
# (e.g., in GNS3/pfSense) then a nested VLAN1-tagged frame traverses
```

DHCP starvation & rogue DHCP

Flood DHCP discoveries → exhaust the pool → clients cannot get an IP
Deploy your own rogue DHCP server → hand out you-as-gateway → easy MITM

```
yersinia dhcp -test
```

STP abuse

Send Bridge Protocol Data Units claiming you are root bridge → network reconverges through you (timing/DoS, MITM potential)

Defenses (the switch stack)

Control	Stops
Port security / MAC limits	Cam overflow
BPDU guard	STP abuse on access ports
PortFast + root guard	STP misroot
DHCP snooping + trusted ports	rogue DHCP
Dynamic ARP Inspection	ARP spoof
802.1X	The whole game – authenticate the device first
VLAN hygiene	double-tagging (native VLAN changes, no user ports on native/trunk)

Learning point

Switches are NOT a security boundary by default – they're a convenience layer. Trusting "we're switched, so it's safe" is 1998 thinking. Segmentation is only as strong as your *enforcement* of it.

Every L2 attack is essentially 'the switch is gullible'. Harden the switch: authenticate devices, lock ports, verify DHCP/ARP.

[← MITM: ARP Spoofing & DNS Poi...](#)

[Session & Token Hijacking →](#)